Neuromuscular conditions (Peripheral Nerve Injuries)

Prof Isaac Sorinola



Learning Outcomes

By the end of this session, students should be able to:

- Explain the classification of peripheral nerve damage
- Explain the pathophysiology of peripheral nerve damage and recovery
- Explain the management of peripheral nerve damage
- Explain the physical management of peripheral nerve damage including the role of the MDT

Classification of neuromuscular disorders

α-motor neurone pathology

- **Poliomyelitis** *Spinal Muscular Atrophy, Motor Neurone Disease (MND/ALS)*

Nerve root

Spinal nerve

Plexus Brachial plexus

Peripheral nerve pathology

- Peripheral nerve lesion – Common peroneal nerve
- Neuropathy - *Guillain-Barré syndrome, CIDP, Charcot-Marie Tooth disease, Diabetic neuropathy*

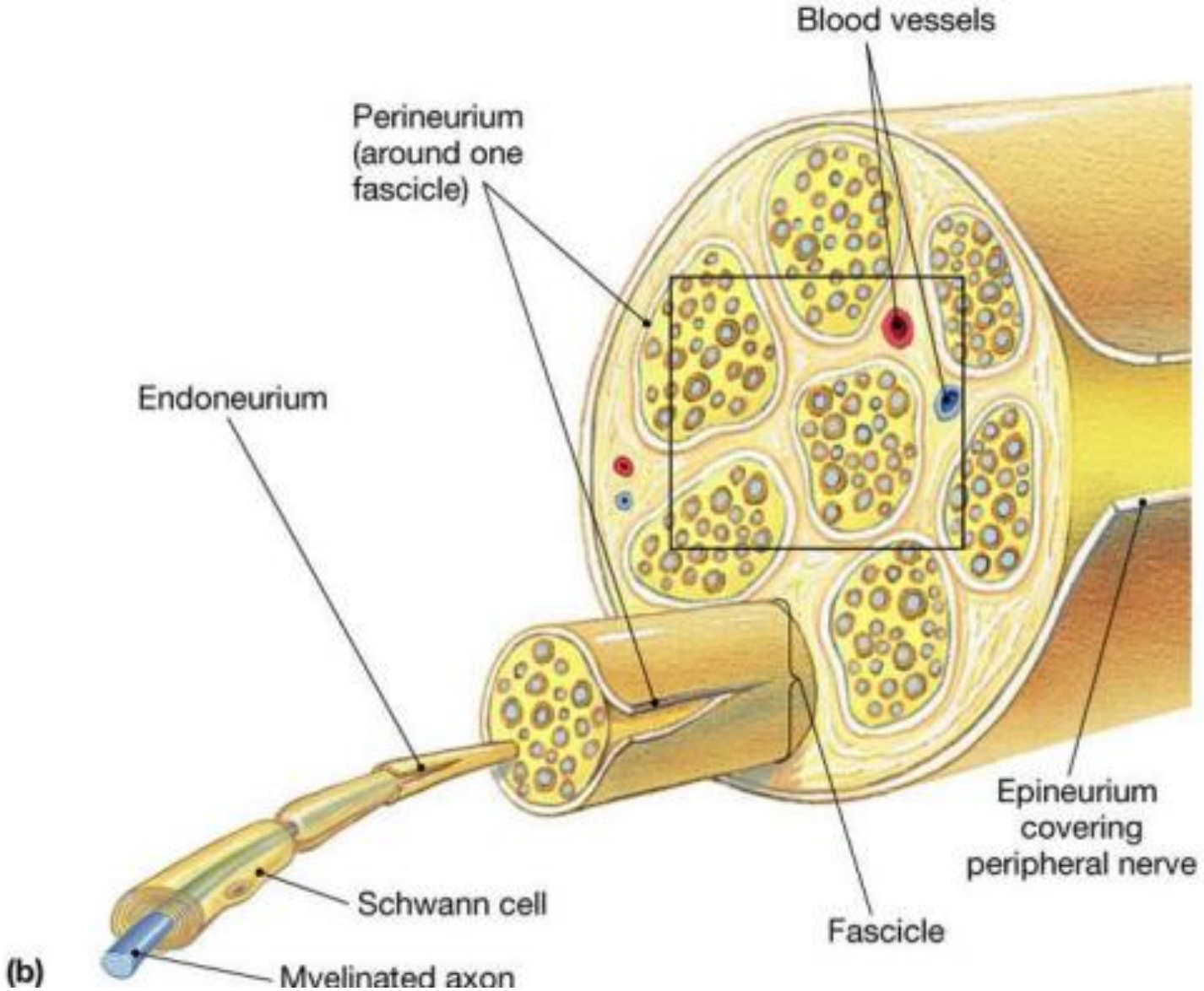
Neuromuscular junction disorders

- *Myasthenia Gravis, Lambert Eaton Myasthenic Syndrome*

Muscle pathology or Myopathy

- *Inflammatory (myositis),*
- *Degenerative (dystrophies).*
- *Contractile (myotonias)*

Structure of a peripheral nerve



Nerve injury classifications – Sunderland (degrees) and Seddon (types)

Neurapraxia (conduction block) 1st degree

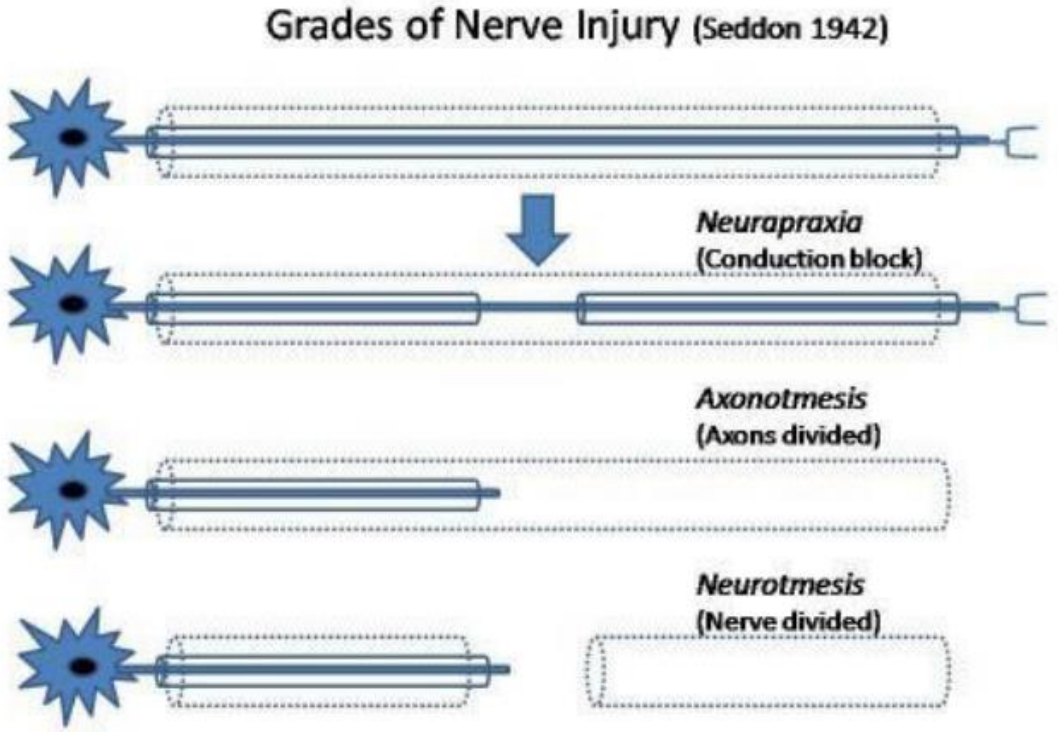
- Mild compression or stretch leading to myelin damage but axon is intact (lasts for 2-12wks)

Axonotmesis 2nd degree

- Arise from stretch injuries resulting to severance of axons but leaving endoneurium (2nd), perineurium (3rd) or epineurium (4th) intact (1mm/day from site of lesion)

Neurotmesis 3rd - 5th degree

- Complete severance of nerves (nerve graft surgery)



Summary of Peripheral Nerve injuries

Types	Cause	Pathology	Signs and Symptoms	Recovery
Neurapraxia (Seddon) First degree (Sunderland)	Prolonged pressure on a nerve	Transient functional loss without affecting loss of nerve continuity	Transient functional loss	Excellent and complete recovery expected in 2 -12 weeks

Summary of Peripheral Nerve injuries

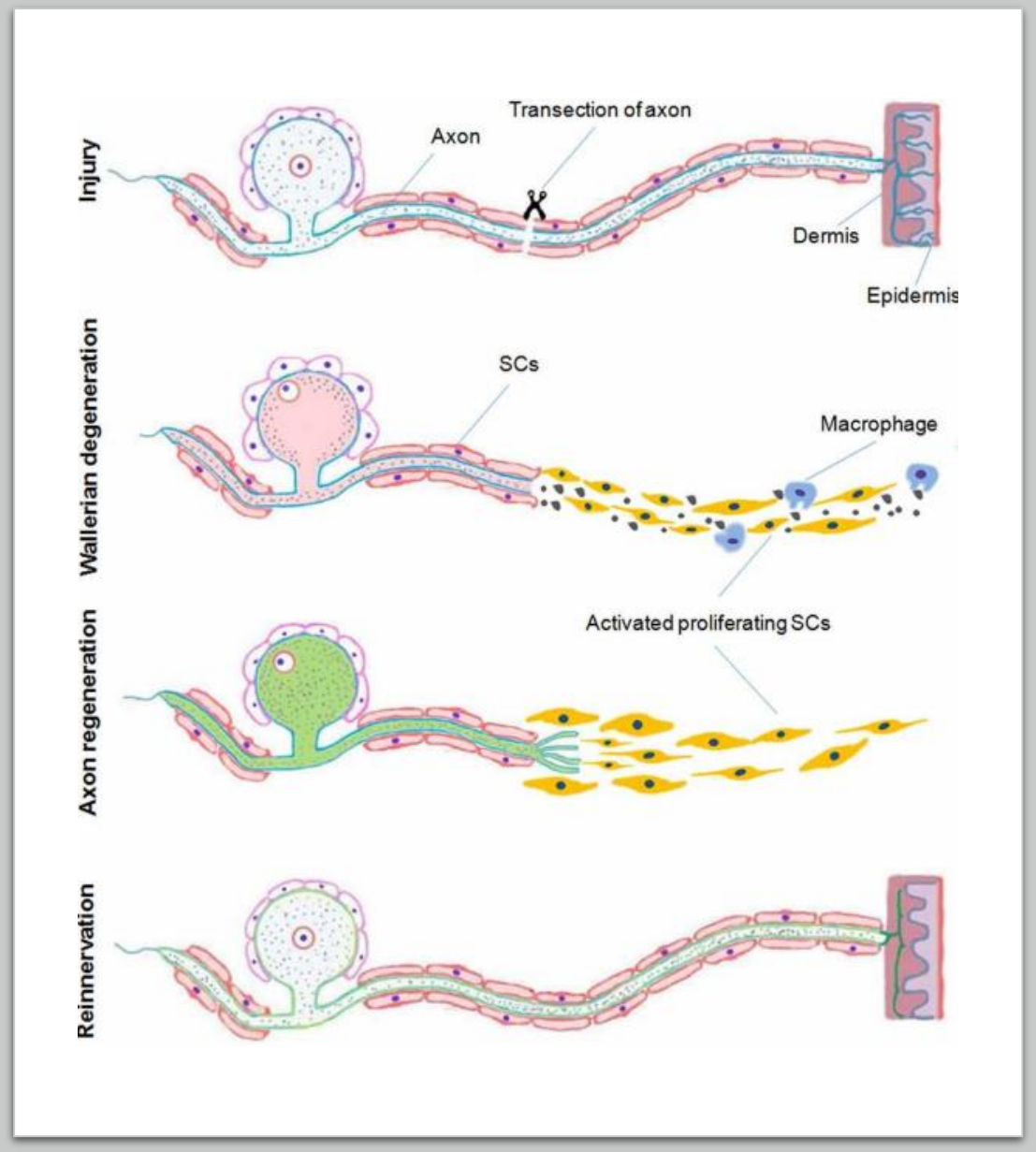
Types	Cause	Pathology	Signs and Symptoms	Recovery
Neurapraxia (Seddon) First degree (Sunderland)	Prolonged pressure on a nerve	Transient functional loss without affecting loss of nerve continuity	Transient functional loss	Excellent and complete recovery expected in 2 -12 weeks
Axonotmesis (Seddon) Second & Third degree (Sunderland)	Stretch injuries on a nerve leading to damage to the nerve axon	Complete disruption of the nerve axon, surrounding myelin but preservation of perineurium and epineurium	Complete loss of function below the injured nerve	Excellent recovery expected – 1mm/day

Summary of Peripheral Nerve injuries

Types	Cause	Pathology	Signs and Symptoms	Recovery
Neurapraxia (Seddon) First degree (Sunderland)	Prolonged pressure on a nerve	Transient functional loss without affecting loss of nerve continuity	Transient functional loss	Excellent and complete recovery expected in 2 -12 weeks
Axonotmesis (Seddon) Second & Third degree (Sunderland)	Stretch injuries on a nerve leading to damage to the nerve axon	Complete disruption of the nerve axon, surrounding myelin but preservation of perineurium and epineurium	Complete loss of function below the injured nerve	Excellent recovery expected – 1mm/day
Neurotmesis (Seddon) Fourth & Fifth degree (Sunderland)	Transection of the nerve by a sharp object or discontinuity of the nerve	Complete functional loss because of disconnection of a nerve.	Complete loss of function below the injured nerve	Variable – 1 mm/day and dependent on the success of nerve repair

Pathophysiology of nerve Injury

- Neurapraxia – no pathological changes or mild changes are seen
- Axonotmesis: Wallerian degeneration occurs distal
- Nuclear migration & chromatolysis in cell body (can lead to apoptosis)
- Wallerian degeneration in axon (distal to lesion) - Calcium mediated process
- Physical fragmentation of both axons and myelin – within hours
- Dissolution and clearance of the myelin sheath and debris (hours to days/weeks)
- Expression of nerve growth factors
- SCs organise into Bands of Bungner to provide pathway for newly sprouting axons during regeneration (approx. 2 months)
- Reinnervation and remyelination

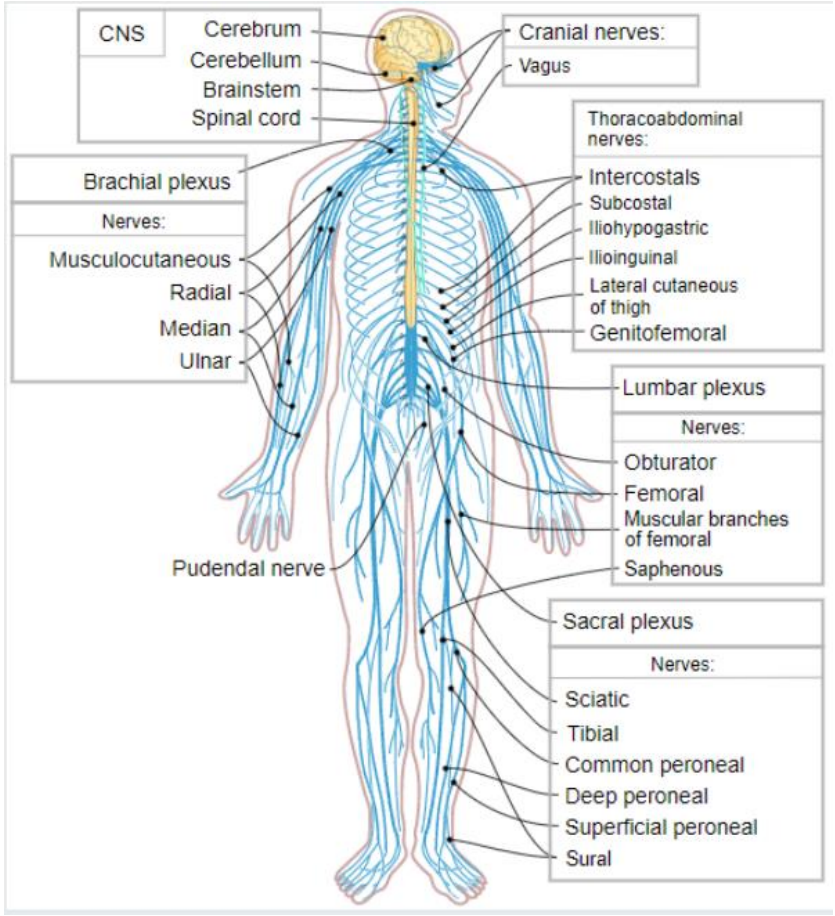


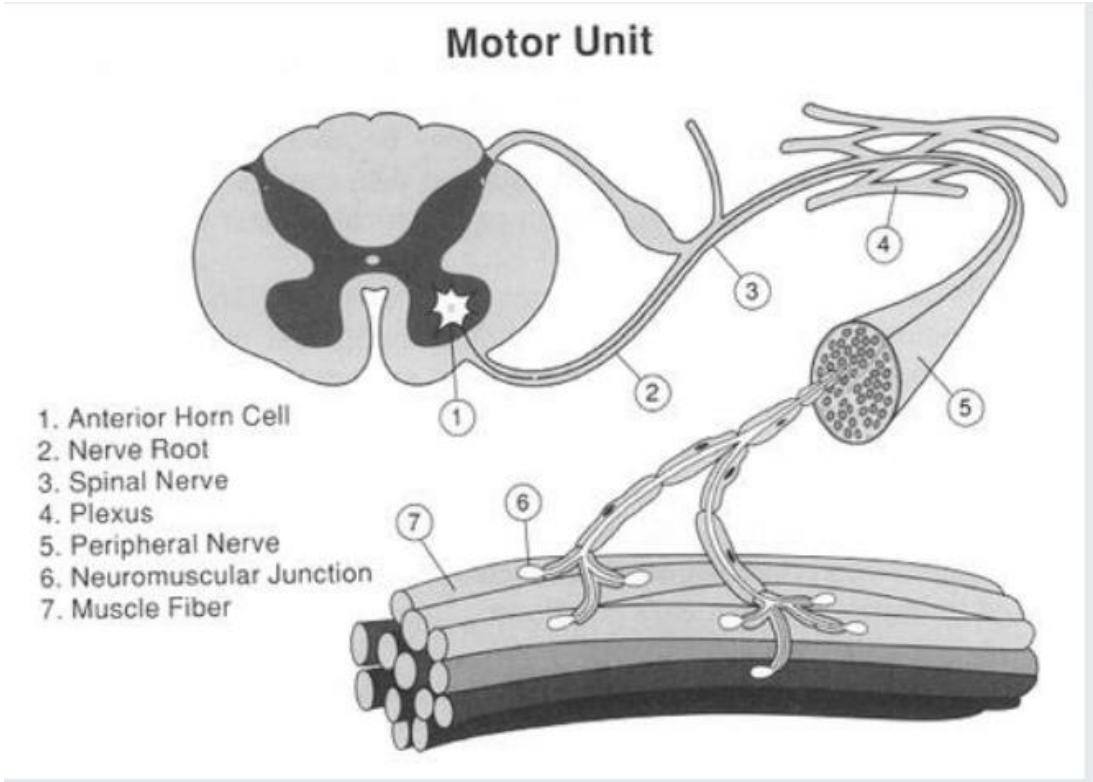
Diagnosing a nerve injury

- Electromyography (EMG)
  - Should be done after 2 – 3 weeks
  - Muscles show denervation changes
  - Complete denervation is characterised by low amplitude sharp waves or fibrillation potential with muscle at rest and absent evoked muscle potential (MUAP)
  - With re-innervation, these changes begin to reverse.

- Nerve Conduction Studies (NCS)
  - Identifying the type and age of peripheral nerve injury
  - In Neurapraxia - compound muscle action potential (CMAP) remains normal distal to the lesion. Drops to zero proximal to the lesion
  - In Axonotmesis – it is present in first week and falls rapidly after Wallerian degeneration

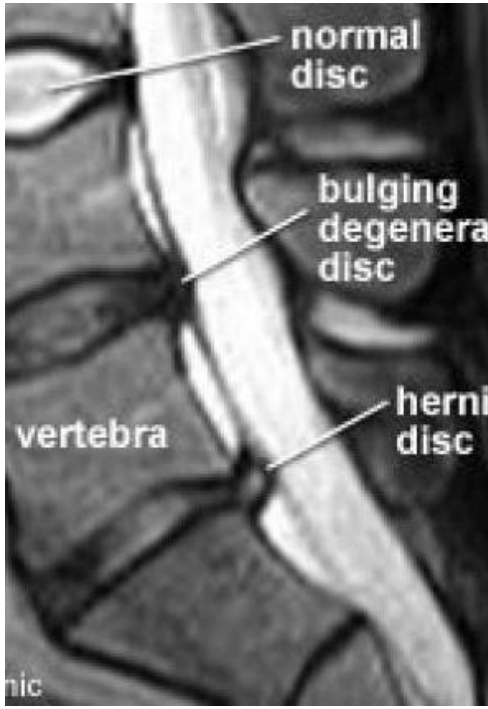
Neuromuscular conditions





1. α-motor neurone,
2. Nerve roots
3. Spinal nerve
4. Plexus
5. Peripheral nerve
6. Neuromuscular junction
7. Muscle fibre

Nerve root disorders (radiculopathy)



Causes

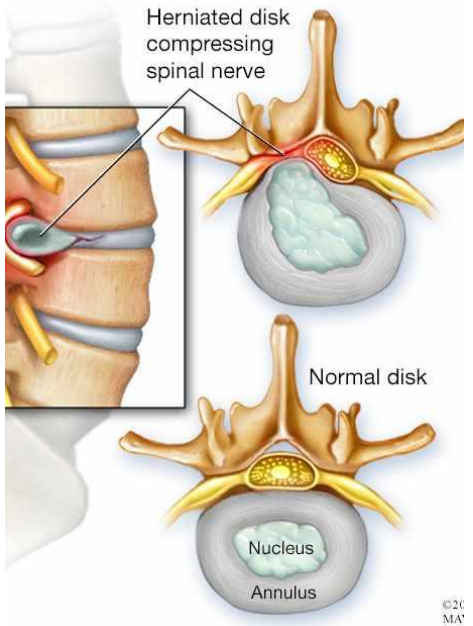
- spondylosis, intervertebral disc disease, trauma, tumours

Symptoms

- Radicular pains: severe, sharp, shooting and/burning pain arising from one or more spinal nerve roots (therefore not necessarily dermatomal) can occur in isolation
- Sensory loss in dermatomal pattern – arises from pathology of either posterior root or spinal nerve

Signs

- Muscle weakness in myotomal distribution – arises from pathology of either anterior root or spinal nerve
- Reduced/absent deep tendon reflexes – arises from pathology of anterior, posterior root or spinal nerve



Symptoms of Common Radiculopathies by Cord Level

Level	Symptoms
C4	Pain in the lower neck and trapezius area with paresthesias involving the lower neck and upper shoulder girdle
C5	Pain in the neck, shoulder, and dorsal forearm with paresthesias and numbness involving the dorsal arm Weakness of the deltoid, biceps, and rotator cuff Decrease in the biceps reflex
C6	Pain in the trapezius ridge and tip of the shoulder, often radiating to the thumb, with paresthesias and sensory impairment in the same areas Weakness of biceps Decreased biceps brachii and brachioradialis reflexes
C7	Pain, paresthesias, and numbness in the shoulder blade and axilla, radiating to the middle finger Weakness of triceps Decreased triceps brachii reflex
T (any)	Bandlike dysesthesias around the thorax (eg, T4 nipple, T10 umbilicus)
L3	Pain, numbness, and paresthesias in the anterior and medial thigh and knee with quadriceps weakness and diminished patellar reflex
L4	Pain, numbness, and paresthesias in the medial lower leg and ankle Weakness of the quadriceps, ankle dorsiflexion, and diminished patellar reflex
L5	Pain in the buttock, posterior lateral thigh, calf, and foot Footdrop with weakness of the anterior tibial, posterior tibial, and peroneal muscles Sensory loss over the anterolateral lower leg and dorsum of the foot
S1	Pain along the posterior aspect of the leg and buttock Weakness of the medial gastrocnemius muscle with impaired ankle plantar flexion Loss of ankle jerk Sensory loss over the lateral calf and foot

Diagnosing Nerve root disorders

- Neuroimaging (MRI or CT scan or myelography)
- Electrodiagnostic testing: Nerve conduction velocity
- Systemic testing for underlying disorders e.g., Diabetes

Damage/disease of the nerve plexus

Causes

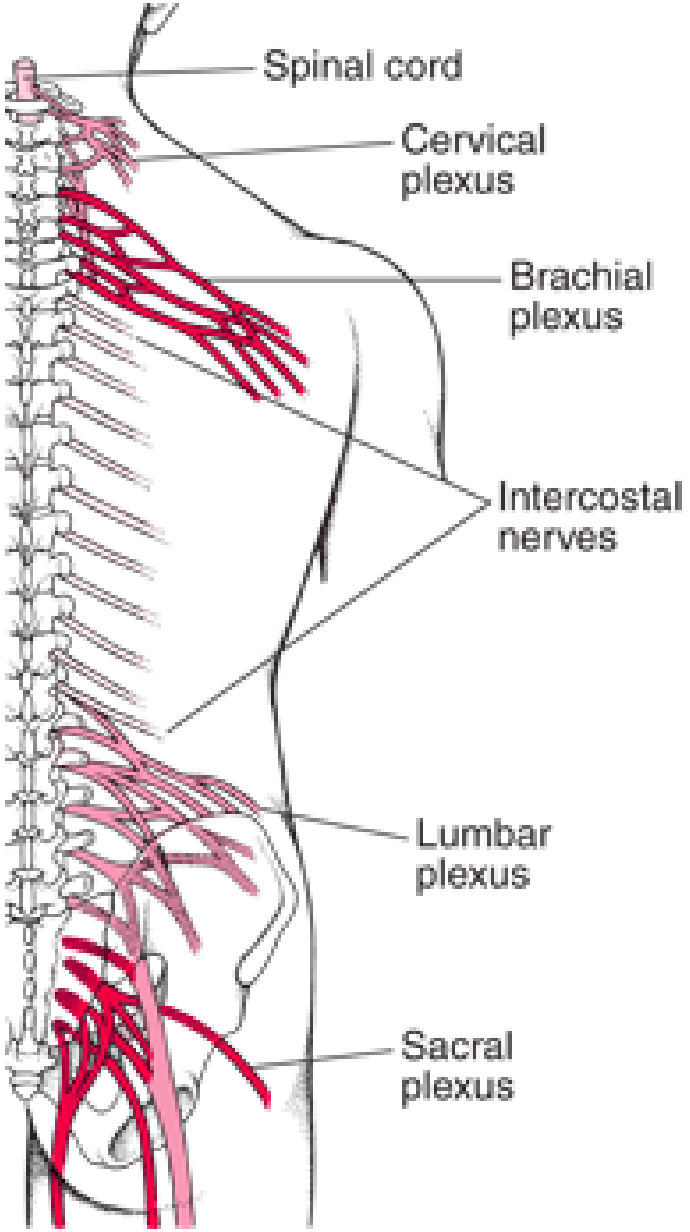
- Trauma
- Malignant infiltration
- Neuralgic amyotrophy
- Radiation therapy
- Compression

Symptoms

- Sensory loss in areas of distribution
- affected e.g. upper versus lower trunk in brachial plexus
- Motor weakness in muscles supplied by plexus

Signs

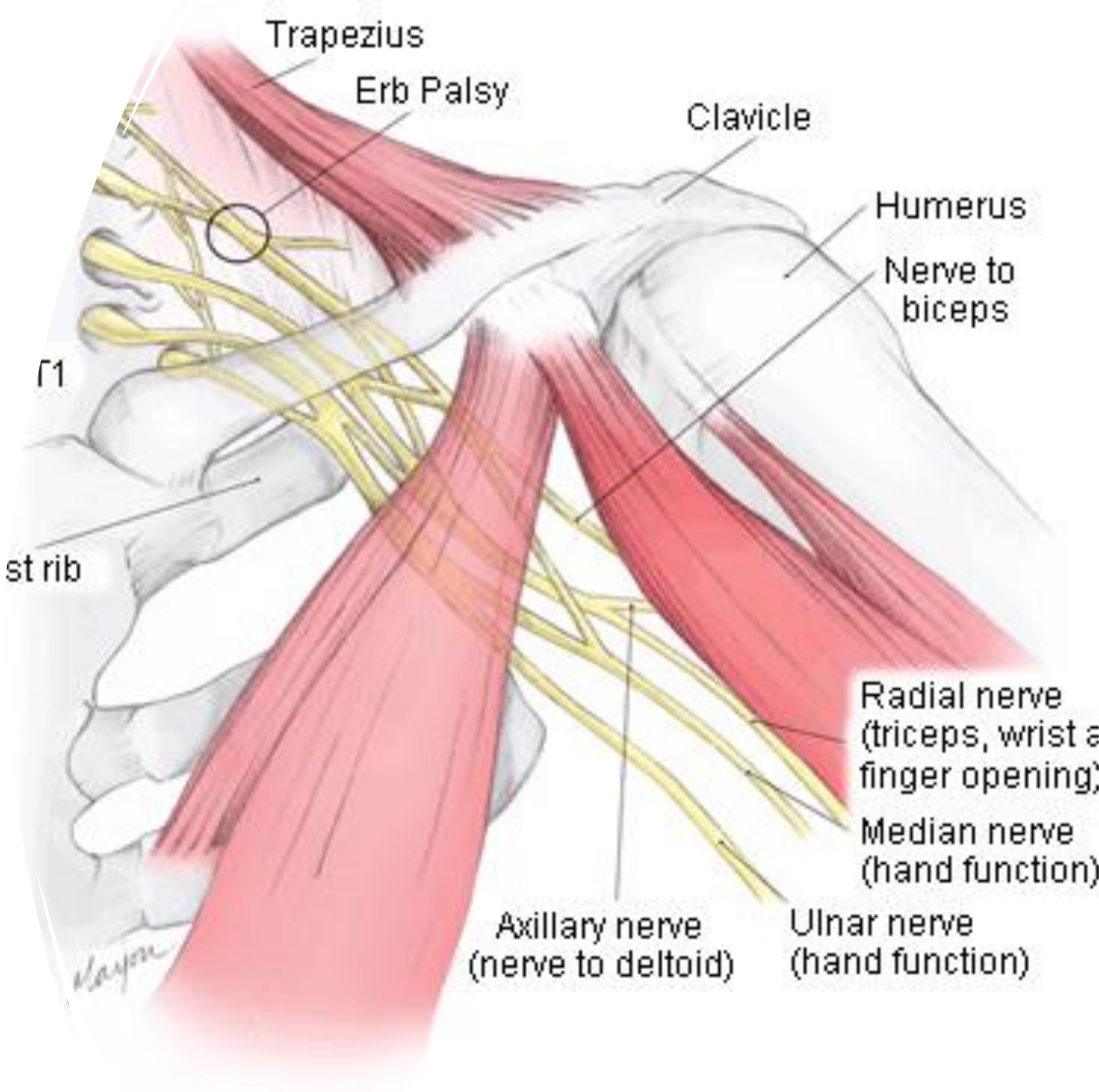
- NCS and EMG – essential for localising lesion especially in brachial plexus



Brachial Plexus injuries

Causes

- Birth injuries or Motorcycle accidents or fall from a height
  - Upper trunk (C5-6) Erb's palsy,
  - Extended Erb's palsy (C5-6 +C7)
  - Lower trunk (C8-T1) - Klumpke's palsy
  - Total (C5-T1) – global plexus palsy





Brachial plexus damage

Symptoms

- Sensory loss - lateral deltoid, arm, and forearm & thumb (C5 & C6 dermatomes)

Signs (C5/6) involvement

- Muscle weakness
  - Shoulder abductors & lateral rotators and elbow flexors
- Atrophy – shoulder and arm
- Deformity (waiter's tip deformity)
- Numbness of the radial digits of the hand and the shoulder
- Triceps, forearm, wrist, and hand strength are typically intact.

Management

- May require surgery to repair or reconstruct

Diagnosing Brachial plexus injury

- Electromyography (EMG)
  - At 6 weeks post injury can show specific pattern of root involvement
  - Can also show signs of recovery if present
  - At 3 months, surgical planning is necessary if no signs of recovery
- Nerve conduction testing
  - To identify the location of the injury
- MRI/CT myelography
  - MRI scanning and myelography of the cervical spine with or without CT imaging has been used to diagnose nerve root avulsion

Examples of brachial plexus injuries

A 24-year-old man presents 6 weeks after a motorcycle accident having sustained polytrauma to the right upper extremity. Three ipsilateral rib fractures are present and humeral and forearm fractures are also noted.

The patient has 0/5 Medical Research Council (MRC) motor scale strength in the entire right upper extremity. There is sensory loss in the C4 through to T1 dermatomes.

There is retention of sensibility in the posterior arm.

A 32-year-old man presents to the emergency department with a history of having fallen from a 10-foot platform at work onto the left shoulder, forcing the head and neck away from the injured shoulder and stretching the neck on the left side.

He has no shoulder or biceps function but retains strong triceps function as well as intact functional strength below the elbow.

He has patchy loss of sensibility over the shoulder and forearm as well as the thumb and index finger.

Management (C5-6) with or without C7 root

1st Line

- Supportive care
- Physiotherapy and occupational therapy
- Pain management
  - Multi-disciplinary team of pain specialist
  - Neuropathic pain
    - gabapentin, carbamazepine, tricyclic antidepressants, topical lidocaine or capsaicin, or opioids (oxycodone)

No recovery 4 - 6 months

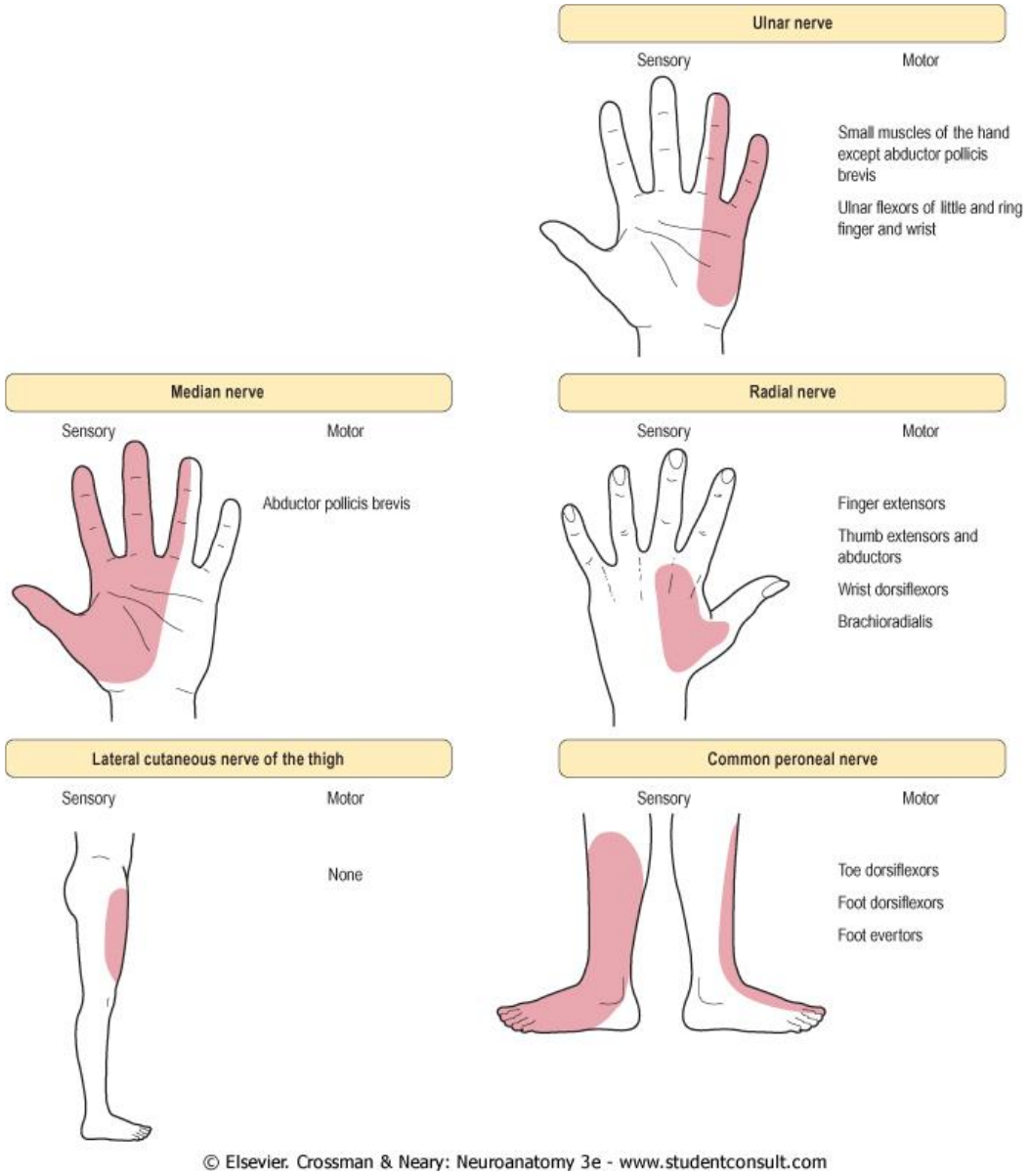
- Primary reconstructive surgery – nerve transfer techniques

No recovery of function by 12 months

- If there is no shoulder function after 12 months – consider shoulder fusion

No arm recovery by 12 months

- Nerve and muscle transfer



Peripheral Nerve injuries

Peripheral nerve lesion (injury):

- laceration,
- severe bruising,
- gunshot,
- traction,
- injection,
- electrical injury

Signs & symptoms

- severe unrelenting pain
- sensory loss & paraesthesia
- Paralysis/muscle weakness with absent/hyporeflexia
- atrophy
- deformity
- autonomic

Assessment

- Subjective and Objective examination
  - Pain assessment
  - Sensory deficit
  - Muscle strength testing
  - Functional deficits
  - Balance deficits
  - Joint ROM
  - Emotional stress

- Common Outcome measures
  - Oxford Scale
  - Visual analogue Scale
  - Relevance functional assessment scale
    - Balance
    - 6-minute peg test
    - 10-meter walk test
    - 6-minute walk test

Surgical Management

Nerve injury	Donor nerve
Musculocutaneous/ upper trunk brachial plexus	Fascicle within ulnar nerve subserving Flexor carpi ulnaris
Axillary nerve/upper trunk brachial plexus	Motor branches of radial nerve to triceps (long or medial head)
Suprascapular nerve	Distal spinal accessory nerve
Posterior interosseous nerve	Branches of median nerve
Axillary nerve	Medial pectoral nerve
Long thoracic nerve/C5 6, 7 injuries	Thoracodorsal nerve
Avulsion of multiple cervical roots	Intercostal nerves
Peroneal nerve	Branches of the tibial nerve (eg, lateral gastrocnemius). May require a nerve graft to extend

Physiotherapy Management

- Pain Management
  - MDT approach – neuropathic pain meds
  - Massage
  - TENS
  - Laser Therapy
  - Acupuncture
- Muscle Weakness
  - Muscle care with E.stim
  - Massage
  - Muscle strengthening
- Functional deficits
  - Motor re-learning
- ROM exercises
- Occupational Therapy - adaptations
- Orthotist – splints/supports



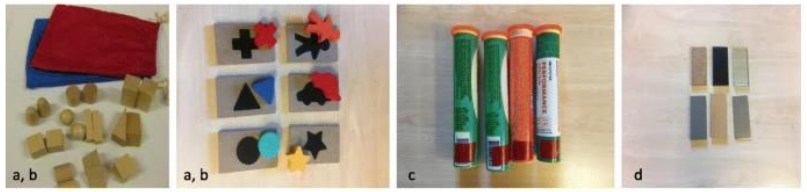


Sensory Re-education

- Touch discrimination for exploring different surfaces and fabrics
- Touch discrimination for identifying objects
- Proprioception for recognising the position of limbs
- Stereognosis



A. Various surfaces and fabrics for touch discrimination



B. Items for touch discrimination of different (a) size; (b) shape; (c) weight and (d) temperature



